

Basic Linux Commands

<code>pwd</code>	Displays current directory	<code>pwd</code>
<code>ls</code>	Lists all files and folders in current directory	<code>ls</code>
<code>-l</code>	Long list with details (size, date, owner, permissions)	<code>ls -l</code>
<code>-a</code>	list files including hidden files (.hiddenFolder)	<code>ls -a</code>
<code>-la / -ll</code>	works like -l but includes hidden files	<code>ls -la</code>
<code>-ls -l <file></code>	show file type, permissions, owner and size for specific file	<code>ls -l <filename></code>
<code>cd <path></code>	Change to specific directory	<code>cd /user</code>
<code>cd ..</code>	Move to parent directory	<code>cd ..</code>
<code>cd ~</code>	Redirect to home directory	<code>cd ~</code>
<code>cd -</code>	Back to previous directory	<code>cd -</code>
<code>cd /</code>	Move to root directory	<code>cd /</code>
<code>mkdir</code>	Create new directory	<code>mkdir <name></code>
<code>rmdir</code>	Remove an empty directory	<code>rmdir <name></code>
<code>tree</code>	Display directory structure as tree	<code>tree</code>
<code>touch <file></code>	Creates new file / update timestamp of existing file	<code>touch text.txt</code>
<code>cp <src> <dest></code>	Copy file to new destination	<code>cp text.txt backup.txt</code>
		<code>cp text.txt</code>
		<code>/home/user/docs</code>
<code>rm <file></code>	Removes file	<code>rm text.txt</code>
<code>rm -r <dir></code>	Removes directory with contents	<code>rm -r user/dir</code>
<code>cat <file></code>	Prints content of file	<code>cat touch.txt</code>
<code>chmod <mode> <file></code>	Change file permissions. 4 = read, 2 = write, 1 = execute 1 st num = owner, 2 nd = group, 3 rd = others +x to run script	<code>chmod 755 script.sh</code>
<code>sudo <com></code>	Run command with admin privileges	<code>sudo apt update</code>

Git Commands

<code>git init</code>	Initialize repository / folder as git	<code>git init</code>
<code>git status</code>	shows the current state of project such as files modified, staged, branch, and untracked files	<code>git status</code>
		<code>-s (short) --show-stash</code>
<code>git clone <url></code>	Clone existing repo using url link	<code>git clone https://github.com/repo</code>
<code>git add</code>	Moves modified files to staging area	<code>git add . (all) git add <filename></code>
<code>git commit</code>	Commits the current staged files	<code>git commit -m "init commit"</code>
<code>git log</code>	Shows all commits made	<code>git log --oneline</code>
<code>git reset</code>	Reverts all changes to state head	<code>git reset --soft --hard</code>
<code>git stash</code>	Saves changes to stashed files to have a clean repository	<code>git stash pop (restore changes) apply show (shows specific stash and shows changes in files) list (lists all stashed items)</code>
		<code>git diff git diff <commit> git diff <filename></code>
<code>git diff</code>	Displays the changes made on different commits and current	
<code>git restore</code>	Removes the changes on unstaged files	
<code>git branch</code>	Show current working branch and lists all branch	<code>git branch git branch <branch-name> (create new branch)</code>
		<code>git checkout <branch-name></code>
<code>git checkout</code>	Switches working branch	<code>git merge branch2 (currently in main)</code>
<code>git merge</code>	Combines the changes from a branch to another	

Three-Tree Architecture of Git

Tree 1: Working Directory -> Working Tree -> the actual place / directory where files are edited

Tree 2: Index -> Staging Area -> files are moved here to be prepared for the next commit

Tree 3: Repository -> HEAD / Git History -> committed project history is stored here

References:

<https://ubuntu.com/server/docs/reference/cli-cheatsheet/>

<https://git-scm.com/cheat-sheet>

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